

SDC310L Project Plan

Name: Holly Hebert

Start Date: January 2, 2026

End Date: January 21, 2026

Overview

This document will be used throughout the course and will serve 3 purposes:

- Define the work breakdown (tasks) you need to complete each week
- Track and report the work you accomplish each week
- Assist in replanning your work to complete in subsequent weeks

Instructions/Submissions

You will submit an updated version of this document each week. The document will serve as a weekly status update to accompany your software code files. Instructions for what information to include in each section of the document are included in the individual section.

Submissions will be as follows:

- Week 1: Initial submission – Due Week 1
 - Complete the “Initial Plan” section of the Work Breakdown table for *each* week of development.
 - Leave all other sections blank
- Week 2: Due Week 2
 - Complete the following sections in the Week 2 Development section:
 - I – Additional Work section of Work Breakdown table
 - II – Work Completed
 - III – Work Remaining
 - IV – Barriers/Roadblocks Encountered
 - V – Requested Assistance
 - Add information to the following sections in the Week 3 Development section:

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- I – Additional Work section of Work Breakdown table
- Week 3: Due Week 3
 - Complete the following sections in the Week 3 Development section:
 - I – Additional Work section of Work Breakdown table
 - II – Work Completed
 - III – Work Remaining
 - IV – Barriers/Roadblocks Encountered
 - V – Requested Assistance
 - Add information to the following sections in the Week 4 Development section:
 - I – Additional Work section of Work Breakdown table
- Week 4: Due Week 4
 - Complete the following sections in the Week 4 Development section:
 - I – Additional Work section of Work Breakdown table
 - II – Work Completed
 - III – Work Remaining
 - IV – Barriers/Roadblocks Encountered
 - V – Requested Assistance
 - Add information to the following sections in the Week 5 Development section:
 - I – Additional Work section of Work Breakdown table
- Week 5: Due Week 5
 - Complete the following sections in the Week 5 Development section:
 - I – Additional Work section of Work Breakdown table
 - II – Work Completed
 - III – Work Remaining
 - IV – Barriers/Roadblocks Encountered
 - V – Lessons Learned

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Week 2 Development: Creating the Database and Application Framework

This week the focus will be on beginning your application development by creating the database your application will use and creating the application infrastructure. Tasks should define table(s) to be created, contents of those tables, and the pages/files you will create as part of the infrastructure. Make sure to refer to the Course Project Overview for the project requirements.

I. Work Breakdown

Add tasks to the table below that break up the work to be completed for the week into smaller, more manageable work items. The Task Name should be a short identifier for the task. The Task Description should include enough detail to understand what needs to be done. Replace the examples in the table with your tasks. Planned Completion is the planned completion date for the task.

Task Name	Task Description	Planned Completion
Initial Plan		
Design Database Schema	Identify required database tables (products, cart items) and define fields, data types, and relationships needed to support the online store.	January 2 nd
Create Database	Create the application database using MySQL and implement the defined tables.	January 2 nd
Populate Product Data	Insert at least seven sample products into the database with required attributes (ID, name, description, cost).	January 2 nd
Set Up Application Framework	Create the initial PHP file structure including configuration, database connection file, and basic page routing.	January 2 nd
Initialize GitHub Repository	Create and configure a GitHub repository and push the initial project	January 2 nd

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	structure and database scripts.	
Additional Work - work not planned, but that needed to be done		
Adjusted database column definitions to resolve MariaDB/phpMyAdmin syntax issues.		
Updated product cost column to support decimal currency values.		
Added a unique composite index to prevent duplicate cart entries for the same session and product.		
Exported the finalized database schema as an SQL file for version control.		

II. Work Completed

From the table above, list the Task Name and the Actual Completion Date for the task.

Task Name	Actual Completion
Design Database Schema	January 2 nd
Create Database	January 2 nd
Populate Product Data	January 2 nd
Set Up Application Framework	January 2 nd
Initialize GitHub Repository	January 2 nd

III. Work Remaining (copy this to Week 3 Additional Work)

From the Initial Plan table above, list the Task Name for any tasks that were planned but not completed.

Task Name
Apply MVC framework to separate application logic, data access, and presentation
Implement catalog and shopping cart functionality using PHP
Add cart navigation and checkout behavior

IV. Barriers/Roadblocks Encountered

Describe any issues you encountered during the week; focus on anything that prevents you from completing all planned work.

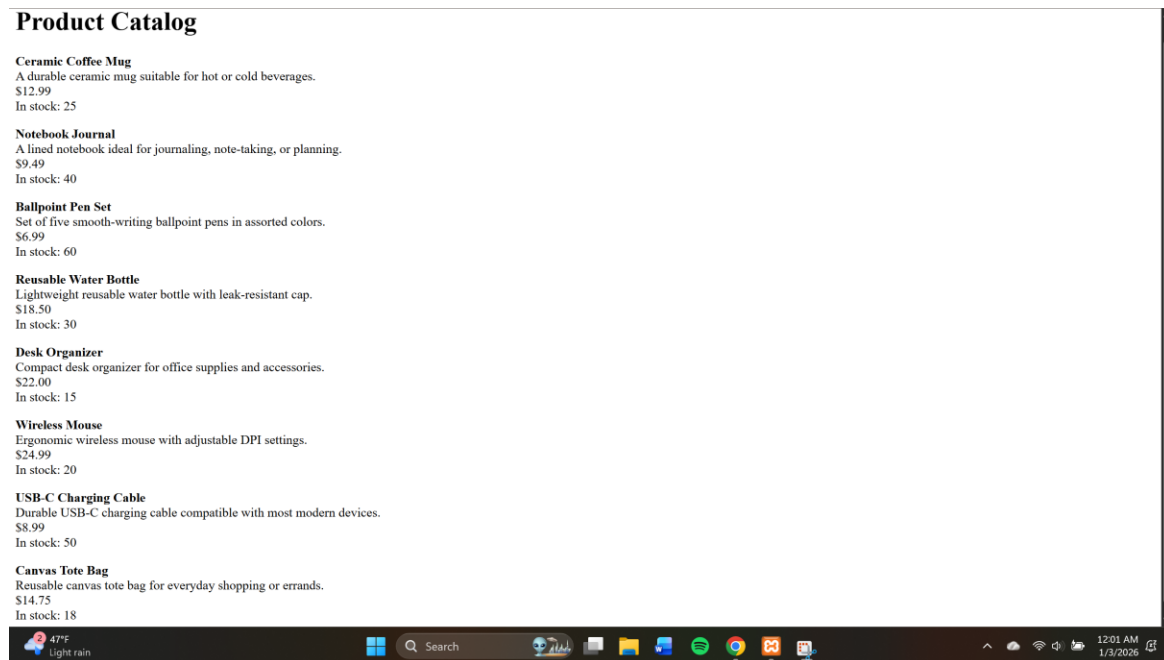
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Some challenges were encountered related to MariaDB and phpMyAdmin behavior, including SQL syntax errors and limitations in setting default values through the graphical interface. These issues require adjustments to data types and column attributes to ensure compatibility. Additionally, becoming familiar with PHP database connections requires extra time and testing.

V. Requested Assistance

Describe any assistance you need to remove barriers/roadblocks that are preventing you from completing all planned work.

No additional assistance is required at this time. The identified issues were resolved through troubleshooting and testing, and development is progressing as planned.



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Week 3 Development: Adding Database Support

This week the focus will be on adding database support to your web application. Support should include database connection processing and all CRUD (Create, Read, Update, Delete) operations you believe will be needed to support your application. Make sure to refer to the Course Project Overview for the project requirements.

I. Work Breakdown

Add tasks to the table below that break up the work to be completed for the week into smaller, more manageable work items. The Task Name should be a short identifier for the task. The Task Description should include enough detail to understand what needs to be done. Replace the examples in the table with your tasks. Planned Completion is the planned completion date for the task.

Task Name	Task Description	Planned Completion
Initial Plan		
Implement Database Connection	Write PHP code to securely connect to the database using appropriate credentials and error handling.	January 8, 2026
Retrieve Product Catalog	Develop PHP logic to retrieve product data from the database and display it in a textual catalog format.	January 8, 2026
Implement Cart Logic	Create PHP logic to add products to the cart, remove products, and update quantities while preventing negative values.	January 8, 2026
Calculate Order Totals	Implement calculations for item totals, tax (5%), shipping (10%), and overall order total.	January 8, 2026
Create Cart Page	Develop a shopping cart page displaying selected items, quantities, individual costs, and totals.	January 8, 2026

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Additional Work - work not completed as part of previous week & unplanned work that had to be completed
Refined database schema usage to support cart operations without modifying the existing database structure
Added session handling to ensure cart persistence across page loads
Tested CRUD functionality end-to-end using local development server (XAMPP)

II. Work Completed

From the table above, list the Task Name and the Actual Completion Date for the task.

Task Name	Actual Completion
Implement Database Connection	January 9, 2026
Retrieve Product Catalog	January 9, 2026
Implement Cart Logic	January 9, 2026
Calculate Order Totals	January 9, 2026
Create Cart Page	January 9, 2026

III. Work Remaining (copy this to Week 4 Additional Work)

From the Initial Plan table above, list the Task Name for any tasks that were planned but not completed.

Task Name
None!

IV. Barriers/Roadblocks Encountered

Describe any issues you encountered during the week; focus on anything that prevents you from completing all planned work.

While working on Week 3, the primary challenge was becoming familiar with PHP syntax and best practices for database interaction, particularly around prepared statements, session handling, and coordinating multiple PHP files. Additional time was spent debugging minor syntax errors and ensuring correct data flow between forms,

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action handlers, and database functions.

V. Requested Assistance

Describe any assistance you need to remove barriers/roadblocks that are preventing you from completing all planned work.

At this time, no additional assistance is required. All planned tasks for Week 3 were completed successfully, and the application is functioning as expected. Future assistance may be requested as additional features are introduced in upcoming weeks.

Your Cart

[← Continue Shopping](#)

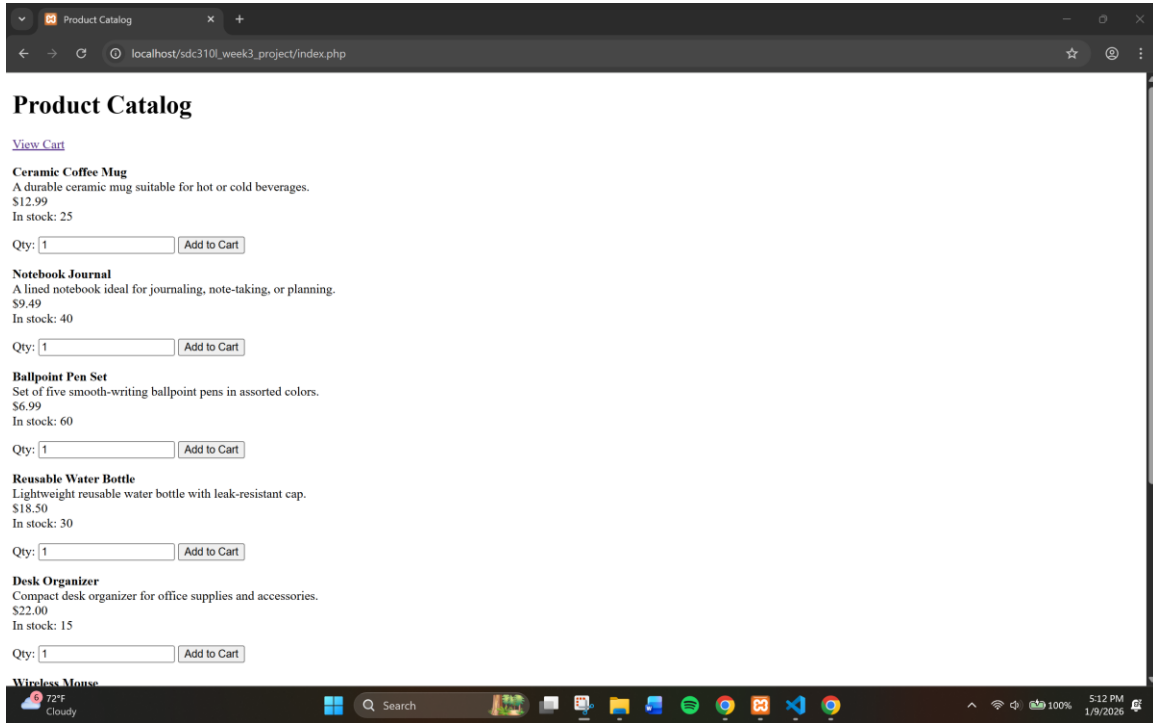
Product ID	Product Name	Quantity	Cost (each)	Line Total	Actions
1	Ceramic Coffee Mug	1 Update	\$12.99	\$12.99	Remove
2	Notebook Journal	2 Update	\$9.49	\$18.98	Remove
7	USB-C Charging Cable	2 Update	\$8.99	\$17.98	Remove

Order Summary

- Total of items ordered (subtotal): \$49.95
- Tax (5%): \$2.50
- Shipping & Handling (10%): \$5.00
- **Order Total:** \$57.44

[Check Out](#)

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Week 4 Development: Architecture – Applying MVC

This week the focus will be on re-architecting the web application to use the MVC architecture. Your tasks should describe your approach to the re-architecture (re-architect in-place or by starting a new project). Make sure to refer to the Course Project Overview for the project requirements.

I. Work Breakdown

Add tasks to the table below that break up the work to be completed for the week into smaller, more manageable work items. The Task Name should be a short identifier for the task. The Task Description should include enough detail to understand what needs to be done. Replace the examples in the table with your tasks. Planned Completion is the planned completion date for the task.

Task Name	Task Description	Planned Completion
Initial Plan		
Plan MVC Structure	Define the separation of concerns between Models, Views, and Controllers for the application.	January 15 th
Refactor Models	Move database access and business logic into Model classes.	January 15 th
Refactor Controllers	Create Controller logic to handle user actions such as adding to cart, updating quantities, and checkout.	January 15 th
Refactor Views	Update PHP/HTML files to serve as Views, focusing only on presentation and output.	January 15 th
Test MVC Integration	Verify that the re-architected application functions correctly after MVC refactoring.	January 15 th
Additional Work - work not completed as part of previous week & unplanned		

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work that had to be completed
Switch database connector from PDO to mysqli to match course/professor requirements while refactoring into MVC.
Fix routing/view path issues (case sensitivity, folder naming like catalog vs catalogue, incorrect required paths).
Remove duplicate/old code paths after MVC refactor (eliminating repeated output and keeping Views clean).

II. Work Completed

From the table above, list the Task Name and the Actual Completion Date for the task.

Task Name	Actual Completion
Plan MVC Structure	January 15 ^h
Refactor Models	January 16 th
Refactor Controllers	January 16 th
Refactor Views	January 16 th
Test MVC Integration	January 16 th

III. Work Remaining (copy this to Week 5 Additional Work)

From the Initial Plan table above, list the Task Name for any tasks that were planned but not completed.

Task Name
None.

IV. Barriers/Roadblocks Encountered

Describe any issues you encountered during the week; focus on anything that prevents you from completing all planned work.

During refactoring, path and file naming issues caused runtime errors (missing required files, view path mismatches, and folder spelling/case differences). Some duplicate logic temporarily existed while transitioning from the previous structure to MVC, which caused repeated output until the old code paths were removed. Additionally, switching

[illegible]

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Week 5 Development: Finalizing the Application

This week the focus will be on finalizing your application development and testing your application. Make sure to refer to the Course Project Overview for the project requirements.

I. Work Breakdown

Add tasks to the table below that break up the work to be completed for the week into smaller, more manageable work items. The Task Name should be a short identifier for the task. The Task Description should include enough detail to understand what needs to be done. Replace the examples in the table with your tasks. Planned Completion is the planned completion date for the task.

Task Name	Task Description	Planned Completion
Initial Plan		
Final Functional Testing	Test all application features including cart updates, calculations, navigation, and checkout behavior.	January 21, 2026
Fix Bugs and Refine UI	Address any identified issues and improve usability and layout consistency.	January 21, 2026
Prepare Test Description Document	Create a document describing test cases, expected results, and actual outcomes.	January 21, 2026
Write Retrospective Document	Reflect on the development process, challenges encountered, and lessons learned.	January 21, 2026
Final GitHub Submission	Ensure all files are committed and pushed to GitHub for final project submission.	January 21, 2026
Additional Work - work not completed as part of previous week & unplanned		

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work that had to be completed
Re-exported the database SQL file to include both structure and seed data after discovering the original export did not include INSERT statements.
Verified database setup instructions to ensure the application runs correctly in a fresh environment.
UI Styling (Catalog + Cart) — Added shared styling via style.css and updated views to match a cohesive interface.
Image Integration — Added product images and dynamic image loading by product_id for catalog and cart.
Asset Organization — Created/confirmed /images folder and ensured paths resolve correctly in MVC views.
Regression Check After UI Changes — Re-tested cart behavior after updating views and routing links.

II. Work Completed

From the table above, list the Task Name and the Actual Completion Date for the task.

Task Name	Actual Completion
Final Functional Testing	January 19, 2026
Fix Bugs and Refine UI	January 19, 2026
Prepare Test Description Document	January 19, 2026
Write Retrospective Document	January 19, 2026
Final GitHub Submission	January 19, 2026

III. Work Not Completed

From the Initial Plan table above, list the Task Name for any tasks that were planned but not completed.

Task Name
None. All planned tasks were completed as scheduled.

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IV. Barriers/Roadblocks Encountered

Describe any issues you encountered during the week; focus on anything that prevents you from completing all planned work.

The primary challenge this week was ensuring that the SQL export included both the database structure and seed data, so the application would run correctly in another environment. This required re-exporting the database using custom export settings and verifying the contents of the SQL file. Otherwise, no major functional issues were encountered during final testing.

V. Lessons Learned

Describe any lessons you learned through the planning and project execution process and anything you would do differently if you were starting over.

This project reinforced the importance of planning, incremental development, and verifying assumptions—especially when preparing a project for use by someone else. Refactoring the application into MVC architecture improved code organization and made the project easier to understand and maintain. If starting over, more attention would be paid earlier to deployment and portability considerations, such as database exports and environment setup.